

Multi-Layer Dos Detection and Severity Scoring System Using
Hybrid ML and DL Approaches.

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Denial of Service (DoS) attacks remain a major threat to modern cybersecurity, disrupting critical network infrastructure through volumetric and protocol-based attacks at the network and transport layers. Most existing intrusion detection systems treat DoS detection as a binary task, classifying traffic only as attack or benign, without measuring the severity or operational impact of different attack types. This limits effective incident response, resource allocation, and mitigation in Security Operations Centers (SOCs).

This research proposes a Multi-Layer DoS Detection and Severity Scoring System that combines Machine Learning (ML) and Deep Learning (DL) techniques to detect four DoS attack types: ICMP flood, SYN flood, IP fragmentation attack, and UDP flood. The system consists of four dedicated components, each assigned a severity weight: ICMP (1.5), SYN (3.0), Fragmentation (2.0), and UDP (2.5). Detection is performed using Random Forest, SVM, CNN, and LSTM models to capture both structured and temporal traffic patterns.

The key contribution is a Severity Scoring Engine that calculates attack impact using $\text{Score} = \text{Attack_Packet_Count} \times \text{Component_Weight}$. Scores are then mapped into four severity levels: Low, Medium, High, and Critical, enabling prioritized mitigation. The system will be trained and evaluated using the CICDDoS2019 and CAIDA datasets, with performance measured through accuracy, precision, recall, F1-score, and AUC-ROC.

By moving beyond binary detection, the proposed system enables SOC analysts to prioritize responses based on impact level rather than simple attack presence. The research supports SDG 9 through resilient digital infrastructure and SDG 16 by strengthening cyber resilience in institutional and government networks.

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We acknowledge the research community whose published works in machine learning-based intrusion detection and DoS attack analysis provided the theoretical foundation for this research. Finally, we thank our families and friends for their unwavering encouragement and moral support throughout this project.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
DoS	Denial of Service
DDoS	Distributed Denial of Service
IDS	Intrusion Detection System
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
SVM	Support Vector Machine
RF	Random Forest
ICMP	Internet Control Message Protocol
SYN	Synchronise (TCP flag)
UDP	User Datagram Protocol
IP	Internet Protocol
SOC	Security Operations Centre
PCAP	Packet Capture
RFE	Recursive Feature Elimination
SVM	Support Vector Machine
AUC-ROC	Area Under the Curve – Receiver Operating Characteristic

INTRODUCTION

Background and Context

Denial of Service (DoS) attacks remain one of the most persistent and disruptive threats in modern cybersecurity. These attacks operate by overwhelming target systems with malicious traffic, rendering critical services unavailable to legitimate users. Organizations across sectors including cloud service providers, financial institutions, healthcare systems, internet service providers (ISPs), and government networks are primary targets due to their reliance on continuous network availability [1] [2].

According to the ENISA Threat Landscape 2024 report, DoS and DDoS attacks continue to rank among the top threats globally, with a notable increase in multi-vector attacks targeting both network and transport layers simultaneously. The report documents that high-profile attack campaigns now commonly employ blended attack patterns combining ICMP floods, SYN floods, UDP floods, and fragmentation-based attacks to maximize service disruption.

Technically, unresolved DoS attacks cause service outages, degraded performance, and resource exhaustion across Layer 3 (Network) and Layer 4 (Transport) protocols. The affected protocols include ICMP (Internet Control Message Protocol) at Layer 3, and TCP and UDP at Layer 4. Economically, prolonged downtime results in significant financial losses, Service Level Agreement (SLA) violations, and reputational damage. Socially, disruptions in critical services such as healthcare and banking systems negatively affect public trust and welfare. From a legal and ethical perspective, failure to detect and respond to high-severity attacks may result in regulatory non-compliance and negligence claims, particularly in critical infrastructure sectors [3] [4].

Despite advances in machine learning and deep learning for intrusion detection, current DoS detection systems predominantly operate on binary outcomes — classifying traffic as either attack or normal. This binary approach fails to differentiate the relative severity of different attack types, leading to inefficient prioritization of mitigation actions. For example, a SYN flood attack that exhausts TCP connection state tables is far more operationally disruptive than an ICMP flood, yet most detection systems treat them identically in terms of response urgency.

This research addresses a clearly identified gap in the literature: the absence of a weighted, multi-layer severity scoring mechanism that enables Security Operations Centre (SOC) analysts and network administrators to make risk-informed decisions based on the actual impact level of detected DoS attacks. The proposed system integrates hybrid ML and DL detection models with an impact-based severity scoring engine across four attack types: ICMP flood, SYN flood, UDP flood, and fragmentation attacks.

This research aligns with SDG 9 (Industry, Innovation, and Infrastructure) by contributing to resilient and secure digital infrastructure, and SDG 16 (Peace, Justice, and Strong Institutions) by enhancing cyber resilience against disruptive threats in institutional and government networks [5].

1.2 Literature Review

A comprehensive review of existing research reveals significant advances in ML and DL-based DoS detection, alongside notable gaps in severity quantification. The following subsections organize findings by theme.

1.2.1 Traditional IDS and Mitigation Systems

Existing DoS mitigation systems largely rely on traditional Intrusion Detection Systems (IDS), load balancers, and rule-based or statistical defenses. Commercial and open-source tools such as Suricata IDS, HAProxy, and Nginx are commonly deployed for volumetric attack mitigation. Romadhon et al. [4] conducted a comparative analysis of HAProxy and Nginx, finding that HAProxy outperforms Nginx in handling high-volume UDP flood attacks due to its event-driven architecture and lower resource consumption. However, these systems focus primarily on mitigation or binary detection and lack intelligent severity differentiation. Karatas, Demir, and Sahingoz [2] surveyed deep learning-based IDS architectures and identified that signature-based systems, despite their good decision time and reliability, cannot detect new or unseen attack types due to their dependency on predefined signatures. Their survey highlights that anomaly-based detection with machine learning and deep learning mechanisms is preferred for dynamic IDS implementation, as it provides the flexibility, robustness, and scalability required to address evolving attack patterns findings that directly motivates the hybrid ML/DL approach adopted in the current research.

1.2.2 Machine Learning Approaches

Sabu et al. [1] proposed a machine learning-based IDS for detecting DoS attacks, demonstrating ensemble techniques including Random Forest, Support Vector Machines (SVM), and AdaBoost achieve high detection accuracy for SYN flood and UDP flood traffic. In the same study, SVM was evaluated using the UNSW-NB15 and CIC-IDS2018 datasets, achieving accuracy of 0.982 and 0.996 respectively through Recursive Feature Elimination (RFE) and Standard Scaler preprocessing, confirming SVM as a robust classifier for DoS intrusion detection. Syafiuddin et al. [3] specifically evaluated AdaBoost for SYN flood and UDP lag attack detection, reporting strong classification performance on benchmark datasets. Tomer and Sharma [6] proposed an ensemble ML model for IoT attack detection combining KNN, Naive Bayes, and Decision Trees with a voting ensemble approach, achieving 99.9% ROC scores on the NSL-KDD dataset while maintaining low execution times suitable for real-time fog-layer deployment. Ramu and Rao [7] evaluated CNN-based detection for TCP/IP, UDP, and ICMP attacks using the KDD99 dataset and demonstrated that CNN significantly outperforms Logistic Regression, MLP, and KNN for ICMP and SYN attack detection.

1.2.3 Deep Learning Approaches

Hernandez et al. [2] proposed a real-time DDoS detection framework for high-speed networks using LSTM, CNN, and hybrid CNN-LSTM architectures, demonstrating superior performance over traditional ML in capturing temporal traffic patterns. Gogoi and Ahmed [9] proposed an LSTM-based deep learning approach specifically for detecting HTTP low and

slow DoS attacks — a category of DoS that does not generate high packet volumes and is therefore invisible to traditional network-layer detection tools. Their approach transforms time-series HTTP request sequences into LSTM input format and classifies them as attack or benign traffic. Experiments on the CIC DoS (2017) dataset augmented with synthetically generated attack traces achieved accuracy and AUC-ROC of 0.99, with false positive and false negative rates both below 0.01. The LSTM architecture employed two stacked LSTM layers with 64 and 32 units respectively, dropout regularization (0.3), and the Adam optimizer with binary cross-entropy loss. This work is directly relevant to the current research as it validates LSTM as a high-performing DL model for sequential DoS traffic pattern classification. Karatas et al [2]. further confirmed in their comparative deep learning IDS survey that deep learning architectures, including LSTM and CNN, outperform traditional machine learning on large-scale intrusion detection datasets and are well-suited for real-time threat classification.

1.2.4 Fragmentation Attack Research

Alyami, Alharbi, and Azzedin [8] conducted a comprehensive survey and simulation of fragmentation attacks on 6LoWPAN IoT networks. Their study categorized fragmentation vulnerabilities including buffer reservation, duplication, alteration, spoofing, and replay attacks. They demonstrated that buffer reservation attacks caused a 67% decrease in throughput and over 50% packet drop ratio in simulated WSN environments. Their reputation-scoring countermeasure improved packet drop ratio by 53.6% with zero additional energy overhead, outperforming prior countermeasures including BRAIN [14% improvement] and SecuPAN [33% increase in energy consumption]. This work validates the severity and operational impact of fragmentation-based DoS attacks and supports the assignment of a medium-high impact weight (2.0) to fragmentation attacks in the proposed severity scoring engine.

1.2.5 IDS Datasets and Evaluation Frameworks

Sabu et al. [1] evaluated their ML-based IDS across three distinct datasets: a self-generated LOIC dataset (accuracy 0.92), UNSW-NB15 (accuracy 0.982), and CIC-IDS2018 (accuracy 0.996). The CIC-IDS2018 dataset, developed by the Canadian Institute for Cybersecurity, contains seven attack categories including DoS, DDoS, brute force, botnet, and web attacks, and is one of the most comprehensive publicly available labelled network traffic datasets. Karatas et al. [13] provided a detailed taxonomy of publicly available datasets for IDS research including KDD Cup99, NSL-KDD, CIC IDS 2017, CSE-CIC-IDS2018, and MCFP Bot Traffic datasets, documenting their characteristics and known shortcomings. The CICDDoS2019 dataset adopted in the current research addresses many of the limitations of earlier datasets by providing labelled L3/L4 attack traffic for ICMP flood, SYN flood, UDP flood, and fragmentation attack categories in a controlled and reproducible environment.

1.2.6 Adaptive Multi-Layer Defense

Sutradhar and Sarkar [5] proposed the A2M (Adaptive Multi-Layered Defense Mechanism) against DDoS attacks, which integrates adaptive mechanisms across multiple network layers.

This architecture demonstrates that multi-layer defenses provide superior resilience compared to single-layer solutions, particularly against multi-vector attacks.

1.3 Research Gap

Despite extensive research into DoS detection using ML and DL techniques, a critical and consistently overlooked gap remains: the absence of a quantitative severity scoring framework that differentiates the relative operational impact of different DoS attack types across multiple network layers.

Current systems treat DoS detection as a binary or multi-class classification problem without quantifying attack severity or impact level. Most systems do not differentiate the relative danger of attack types — for example, SYN flood attacks that exhaust TCP connection state tables versus ICMP floods with comparatively limited protocol impact. This lack of differentiation results in inefficient response prioritization, delayed mitigation decisions, and increased risk to critical services.

Additionally, most existing architectures focus on either detection or mitigation as isolated functions, with no integrated severity-aware decision mechanism bridging the two. Fragmentation attacks in particular are frequently merged with generic DoS categories in detection frameworks, with no dedicated severity quantification methodology.

This research addresses this gap through three key contributions: (1) a multi-layer detection framework specifically targeting ICMP flood, SYN flood, UDP flood, and fragmentation attacks using hybrid ML/DL models; (2) a novel impact-based severity scoring engine that assigns quantitative weights to attack types based on their protocol-level resource exhaustion potential; and (3) a unified severity aggregation mechanism enabling comparative attack prioritizations across simultaneous multi-vector attack scenarios.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a multi-layer DoS detection and severity scoring system that accurately detects ICMP flood, SYN flood, UDP flood, and fragmentation attacks using hybrid ML and DL models, and quantifies their relative operational impact through an impact-based severity scoring engine to enable risk-aware mitigation prioritization

2.2 Specific Objectives

1. To extract and analyze relevant IP fragmentation traffic features including fragment packet rate, datagram tag frequency, incomplete fragment ratios, and time-window reassembly statistics from network traffic datasets (CICDDoS2019 and CAIDA).
2. To implement and evaluate hybrid ML models (Random Forest and SVM) for structured fragmentation traffic feature classification, achieving a target detection accuracy of at least 90%.
3. To implement and evaluate deep learning models (CNN or LSTM) for capturing temporal fragmentation traffic patterns and improving detection performance under high-volume fragmented packet conditions.
4. To develop and validate a weighted severity scoring engine that multiplies the detected fragmentation attack packet count by an impact weight of 2.0 and maps the resulting score to severity levels: Low, Medium, High, and Critical.
5. To generate a structured fragmentation attack severity ranking table that enables SOC analysts to assess and prioritize fragmentation-based DoS attack responses based on computed severity levels.
6. To validate the severity scoring mechanism against attack impact indicators including traffic volume, persistence duration, buffer exhaustion rate, and system resource utilization metrics.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach combined with an Experimental Research methodology. The DSR framework guides the design and construction of the severity-aware DoS detection system as an artefact, while the experimental approach governs systematic evaluation against benchmark datasets. The research follows an iterative development and evaluation cycle: design, implement, experiment, evaluate, and refine.

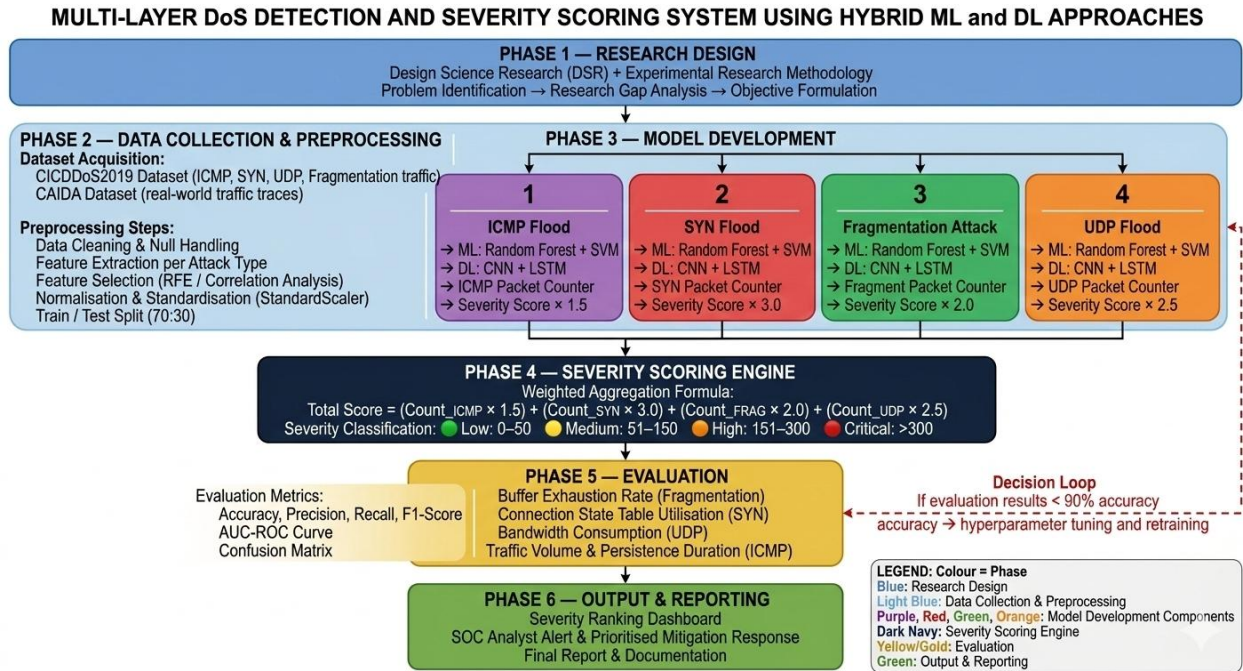


Figure 1 Methodology

3.2 Technical Methods and Algorithms

3.2.1 Multi-Layer Attack Detection

A hybrid detection approach is adopted combining:

- Machine Learning models: Random Forest and Support Vector Machines (SVM) for structured statistical feature classification. These are selected for their interpretability, computational efficiency, and proven performance on network traffic datasets [1] [3].
- Deep Learning models: Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks for capturing temporal and sequential traffic patterns. CNN models detect spatial feature patterns in traffic windows, while LSTM models learn temporal dependencies in sequential packet flows [2] [7].

The exact model configuration will be refined during experimentation based on per-component performance metrics. The hybrid approach leverages the complementary strengths of both model families.

3.2.2 Severity Scoring Engine

The core contribution of this research is the Severity Scoring Engine. Each detected attack type is assigned to an impact-based weight derived from literature, protocol impact analysis, and resource exhaustion potential:

Attack Type	Weight	Protocol Layer	Justification
SYN Flood	3.0	Layer 4 (TCP)	High connection state exhaustion impact
UDP Flood	2.5	Layer 4 (UDP)	High bandwidth and resource consumption
Fragmentation Attack	2.0	Layer 3 (IP)	Buffer exhaustion and reassembly overhead
ICMP Flood	1.5	Layer 3 (ICMP)	Moderate bandwidth impact, lower state exhaustion

Table 1 severity engine

Note: Weights are initial estimates and will be refined through empirical evaluation, attack impact metrics, and expert judgment during project execution.

The total severity score is computed as: $\text{Total Score} = \sum(\text{Attack_Count}_i \times \text{Weight}_i)$ for all detected attack types i . This score is then mapped to severity levels: Low (0-50), Medium (51-150), High (151-300), Critical (>300). Thresholds are indicative and will be calibrated during evaluation.

3.3 Tools and Platforms

- Python 3.x – Primary implementation language for ML/DL model development and data processing.
- Scikit-learn – Random Forest and SVM implementation for ML-based detection components.
- TensorFlow / Keras – CNN and LSTM model implementation for DL-based components.
- Wireshark / PyShark – Network traffic capture and packet-level feature extraction.
- Jupyter Notebook / Anaconda Navigator – Experimentation, visualization, and result analysis.
- CICDDoS2019 Dataset – Primary benchmark dataset containing labelled ICMP flood, SYN flood, UDP flood, and fragmentation attack traffic.

- CAIDA Dataset – Secondary dataset for cross-validation and extended evaluation.

3.4 Data Collection and Ethical Considerations

This project relies exclusively on publicly available, anonymized benchmark datasets (CICDDoS2019 and CAIDA). No real-world attack execution is performed, no live systems are targeted, and no personal or identifiable data is processed. The research strictly adheres to ethical guidelines by focusing on defensive cybersecurity mechanisms. Formal ethical clearance is not required as the project does not involve human participants or sensitive personal data.

3.5 Validation Strategy and Performance Metrics

The detection models will be evaluated using the following performance metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC, False Positive Rate (FPR), and Detection Rate (DR). The severity scoring mechanism will be validated by comparing computed severity levels against attack impact indicators including traffic volume, persistence duration, and system resource utilization. Cross-validation (80:20 train-test split with k-fold cross-validation) will be applied to ensure generalizability.

Metric	Description	Target
Detection Accuracy	Percentage of correctly classified traffic instances	$\geq 90\%$
Precision	Ratio of true attack detections to total predicted attacks	$\geq 88\%$
Recall (Sensitivity)	Ratio of detected attacks to total actual attacks	$\geq 88\%$
F1 Score	Harmonic mean of Precision and Recall	$\geq 88\%$
False Positive Rate	Percentage of normal traffic misclassified as attacks	$\leq 5\%$
Severity Score Accuracy	Alignment of computed severity levels with expert expectations	Cross-validated against literature

Table 2 performance matrix

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Architecture Overview

The proposed system is structured as a prototype-level near real-time detection framework. The architecture follows a component-based pipeline design comprising four parallel detection components, a central aggregation module, and a severity scoring and classification layer.

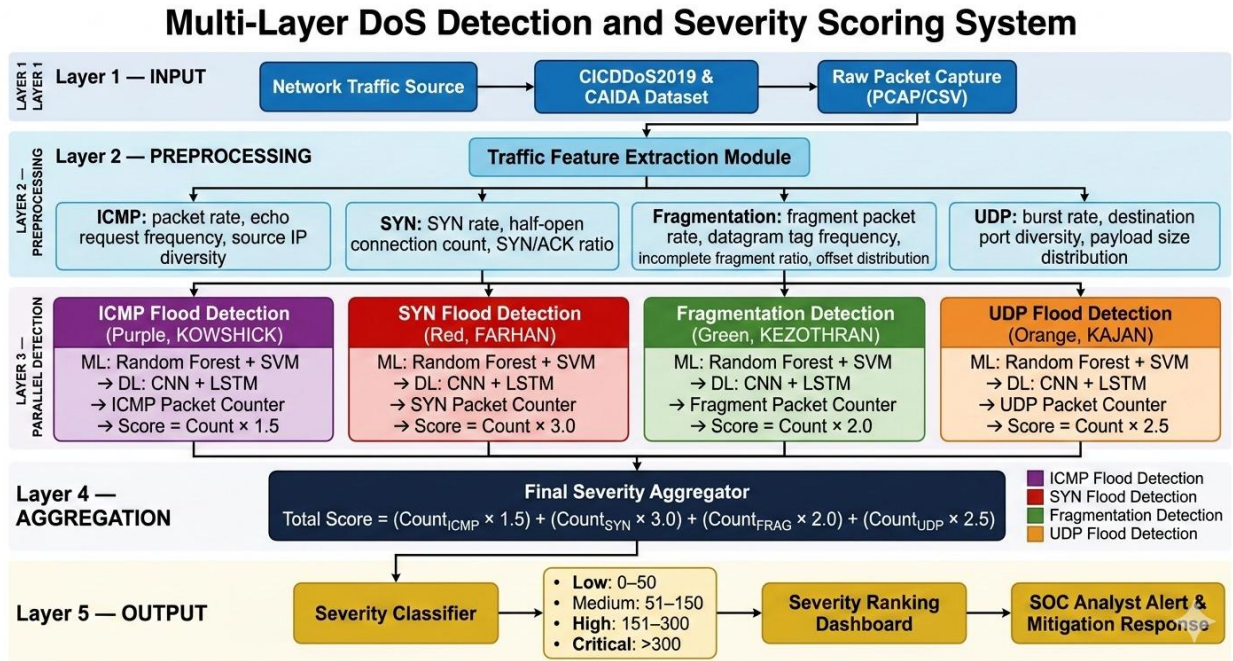


Figure 2 architecture

The system processes network traffic through the following pipeline:

1. Traffic Capture and Feature Extraction – Network traffic (near real-time or offline pcap) is monitored and Layer 3/Layer 4 packet features are extracted using time-window analysis. Features include packet rate, protocol flags, echo request counts, SYN packet counts, UDP burst metrics, and IP fragmentation indicators.
2. Parallel Attack Detection Components – Four independent modules simultaneously analyse traffic for their respective attack types: ICMP Flood (Component 1), SYN Flood (Component 2), Fragmentation Attack (Component 3), and UDP Flood (Component 4). Each component applies dedicated ML/DL models and performs packet counting.
3. Component-Level Severity Scoring – Each component generates an individual severity score by multiplying its detected packet count by its predefined impact weight. A component-specific scoring table is produced.
4. Severity Aggregation – The Final Severity Aggregator collects all component scoring tables and computes the total combined severity score.
5. Severity Classification and Output – The total score is mapped to a severity level (Low, Medium, High, Critical) and presented to security operators alongside detected attack types, individual scores, and mitigation priority guidance

Stage	Component	Description
INPUT	Network Traffic Source	CICDDoS2019 / CAIDA datasets; near real-time or offline PCAP traffic containing IP fragmented packet streams
INPUT	Feature Extraction Module	Extracts fragmentation-specific L3 features including fragment packet rate, datagram tag frequency, incomplete fragment ratio, fragment offset distribution, and time-window reassembly statistics
PROCESS	ML Detection Engine	Random Forest and SVM classifiers for structured fragmentation traffic feature classification
PROCESS	DL Detection Engine	CNN/LSTM networks for temporal fragmentation traffic pattern recognition and incomplete reassembly sequence detection
PROCESS	Fragmentation Packet Counter	Counts detected malicious fragmented IP packets per time window, including buffer reservation attempts and incomplete datagram sequences
PROCESS	Severity Scoring Engine	Applies weight 2.0 to fragmentation packet count; computes individual component severity score
PROCESS	Severity Classifier	Maps computed score to Low / Medium / High / Critical severity levels
OUTPUT	Fragmentation Scoring Table	Structured table with attack type, fragment packet count, severity score, and severity level
OUTPUT	Final Aggregator Feed	Feeds fragmentation of severity score and ranking data to the Combined Scoring Ranking Table

Table 3 process follows

4.2 Component Interactions and Data Flow

Each of the four detection components operates independently on the shared traffic feature stream. Component outputs feed into the central Final Severity Aggregator, which produces a Combined Scoring Ranking Table. This table constitutes the primary decision-support output for SOC analysts. The architecture supports extensibility additional attack type components can be integrated without modifying the core aggregation logic.

4.3 Scalability and Security Considerations

The prototype is designed for experimental evaluation environments. For future production deployment, architecture supports horizontal scaling through parallelization of detection

components. The use of publicly available, anonymized datasets ensures no security or privacy risks during development. All trained model artefacts will be stored securely and access controlled.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- **Primary Users:** SOC analysts and network administrators who require real-time threat awareness and mitigation priority guidance.
- **Secondary Users:** IT security managers who use severity reports for risk assessment and compliance reporting.
- **Organizational Stakeholders:** Cloud service providers, ISPs, financial institutions, healthcare IT departments, and government network teams.
- **Research Stakeholders:** Academic supervisors and research evaluators assessing the system's contribution to cybersecurity knowledge.

5.2 Functional Requirements

1. The system shall capture and process network traffic features from pcap files or near real-time network interfaces.
2. The system shall detect ICMP flood attacks by analyzing ICMP packet rates and echo request patterns.
3. The system shall detect SYN flood attacks by monitoring TCP SYN packet counts and connection attempt rates.
4. The system shall detect IP fragmentation attacks by identifying and counting fragmented IP packets.
5. The system shall detect UDP flood attacks by analysing UDP packet burst rates and volume.
6. The system shall apply predefined impact-based weights to each detected attack type to compute individual severity scores.
7. The system shall aggregate individual severity scores into a total combined severity score.
8. The system shall classify the total severity score into one of four levels: Low, Medium, High, or Critical.
9. The system shall produce a Combined Scoring Ranking Table presenting all detected attacks, their individual scores, and the overall severity level.
10. The system shall support offline evaluation using labelled benchmark datasets (CICDDoS2019, CAIDA).

5.3 Non-Functional Requirements

- **Security:** The system shall process only publicly available or anonymized datasets. No real-world attack execution shall be performed.
- **Performance:** The detection pipeline shall produce severity classification outputs within an acceptable time window suitable for near real-time evaluation (prototype level).
- **Usability:** Output reports shall be presented in a structured, human-readable format comprehensible to SOC analysts without specialized ML expertise.
- **Scalability:** The modular component architecture shall support addition of new attack detection modules without modifying the core aggregation engine.
- **Reliability:** Detection models shall achieve a minimum F1-Score of 0.90 and ROC-AUC of 0.95 on benchmark datasets.
- **Maintainability:** Severity weight parameters shall be configurable and refinable through empirical evaluation without requiring full system retraining.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market comprises organizations that operate mission-critical network infrastructure with continuous availability requirements. Key customer personas include: (1) SOC-as-a-Service providers seeking to enhance threat triage automation; (2) mid-to-large ISPs requiring intelligent DDoS severity classification for SLA protection; (3) financial institutions and healthcare IT teams requiring risk-aware security monitoring for regulatory compliance; and (4) cloud security platform vendors seeking differentiating severity intelligence features.

6.2 Value Proposition

The system provides a unique value proposition through its severity-aware detection capability. Unlike binary detection tools, the proposed system enables organizations to prioritize their response to the most damaging attacks in real time, reducing mean time to respond (MTTR) and minimizing operational impact. The weighted severity scoring engine provides quantitative risk intelligence not available in existing open-source or commercial DoS detection tools.

6.3 Revenue Model

- **SaaS Subscription:** Cloud-hosted severity intelligence API for integration into existing SIEM platforms (monthly/annual licensing).
- **Enterprise Licensing:** On-premises deployment for organizations with strict data sovereignty requirements.

- Consulting and Integration Services: Implementation support for custom network environments.

6.4 Competitive Advantage and IP Considerations

The key competitive advantage is the novel severity scoring mechanism and its integration with multi-layer hybrid ML/DL detection, a combination not currently available in the open-source or commercial market. The severity weight calibration methodology and aggregation model may be considered for patent protection following empirical validation. Publication of peer-reviewed findings will establish academic priority.

7. BUDGET AND JUSTIFICATION

7. Total Estimated Budget: LKR 150,000

Item	Estimated Cost (LKR)	Justification
Cloud Computing (AWS/GCP – Model Training)	25,000	GPU-accelerated training of CNN/LSTM models on CICDDoS2019 dataset
Software Licenses	0	TensorFlow, Scikit-learn, Wireshark (open source – minimal cost)
Dataset Access and Storage	5,000	Storage infrastructure for CICDDoS2019 and CAIDA datasets
Hardware (Lab equipment, additional RAM)	0	Local experimentation environment setup
Human Effort	0	Time allocation for implementation, testing, and documentation
Miscellaneous (printing, communications)	5,000	Report printing, conference fees, communication costs
Total	35,000	

Table 4 Budget

8. WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Assignee	Timeline	Deliverable
T3.1	Literature review on IP fragmentation attacks and ML/DL detection approaches	KEZOTHRAN G	Week 1–3	Annotated bibliography
T3.2	Dataset acquisition and preprocessing (CICDDoS2019, CAIDA)	KEZOTHRAN G	Week 2–4	Preprocessed dataset
T3.3	Fragmentation traffic feature extraction and selection	KEZOTHRAN G	Week 4–6	Feature matrix
T3.4	Random Forest and SVM model implementation for fragmentation detection	KEZOTHRAN G	Week 5–8	Trained ML models
T3.5	CNN/LSTM deep learning model implementation for fragmentation traffic patterns	KEZOTHRAN G	Week 7–10	Trained DL models
T3.6	Fragmented IP packet counting module development	KEZOTHRAN G	Week 9–11	Packet counter module
T3.7	Severity scoring engine implementation (weight = 2.0)	KEZOTHRAN G	Week 10–12	Scoring engine
T3.8	Severity classification and fragmentation attack ranking table generation	KEZOTHRAN G	Week 12–14	Fragmentation Scoring Table
T3.9	Model evaluation and validation	KEZOTHRAN G	Week 13–16	Evaluation report

Task ID	Task Description	Assignee	Timeline	Deliverable
T3.10	Integration with Final Severity Aggregator	All team members	Week 15–17	Integrated component
T3.11	Final documentation and report writing	KEZOTHRAN G	Week 16–18	Final report

Table 5 task

9. GANTT CHART

Task Description	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11	Wk 12	Wk 13	Wk 14	Wk 15	Wk 16	Wk 17	Wk 18
T3.1 Literature Review on Fragmentation Attacks & ML/DL Detection	•	•	•															
T3.2 Dataset Acquisition & Preprocessing (CICDDoS2019, CAIDA)		•	•	•														
T3.3 Fragmentation Feature Extraction & Selection				•	•	•												
T3.4 Random Forest & SVM Model Implementation					•	•	•	•										
T3.5 CNN / LSTM Deep Learning Model Implementation							•	•	•	•								
T3.6 Fragmented IP Packet Counter Module Development								•	•	•								
T3.7 Severity Scoring Engine Implementation (Weight = 2.0)									•	•	•							
T3.8 Severity Classification & Ranking Table Generation											•	•	•					
T3.9 Model Evaluation & Validation												•	•	•	•	•		
T3.10 Integration with Final Severity Aggregator (All Team)															•	•	•	•
T3.11 Final Documentation & Report Writing																•	•	•

Figure 3 Gantt chart

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10.APPENDICES

Appendix A: Risk Analysis

Risk	Likelihood	Impact	Mitigation
Dataset access issues (CICDDoS2019/CAIDA unavailability)	Low	High	Maintain backup datasets (NSL-KDD, UNSW-NB15). Pre-download datasets at project start.
ML/DL model underperformance on specific attack types	Medium	Medium	Iterative model tuning: hybrid ensemble approach reduces single-model failure risk.
Severity weight inaccuracy (initial estimates)	Medium	Medium	Empirical calibration phase built into methodology; weights are explicitly noted as refinable.
Team member availability / workload conflicts	Low	Medium	Clear WBS with independent component ownership. Modular design allows parallel development.
Computational resource limitations for DL model training	Medium	Low	Cloud compute budget allocated Google Colab as fallback.

Table 6 Risk Analysis

Appendix B: Ethical Considerations Statement

This research project does not involve human participants, personal data, or sensitive identifiable information. All experiments are conducted using publicly available, anonymized network traffic datasets (CICDDoS2019, CAIDA, NSL-KDD). No real-world attack execution against live or production systems is performed at any stage. The research is strictly defensive in nature, focusing on detection and severity assessment rather than attack capability development. All dataset usage complies with the respective dataset licensing terms. This research complies with SLIIT institutional research ethics policies and does not require formal ethical clearance.

